Phase 1: Brainstorming & Idea Prioritization

Smart Healthcare Risk & Chronic Disease Surveillance Capstone Project

1. Problem Identification & Ideation

In the modern healthcare domain, non-communicable chronic diseases account for over 70% of hospital admissions and healthcare expenditures.

Brainstorming identified high-leverage intervention targets: cardiovascular risk prediction, metabolic syndrome tracking, multi-morbidity detection, and lifestyle habit quantification.

2. Idea Prioritization Matrix

Initiative Idea	Clinical Impact	Data Feasibility	Analytical Complexity	Priority Rank
Calibrated Health Risk Index	High	Immediate (Cohort data)	Medium (Multi-biomarker)	Priority 1 (Selected)
Interactive Tableau Story Arc	High	High (Cleaned CSV)	Medium (Visual BI)	Priority 1 (Selected)
Real-time Clinical Simulator	High	High (Flask API)	Medium (REST + JS)	Priority 1 (Selected)
Longitudinal Genetic Profiling	Extreme	Low (No genomic markers)	High	Deferred (Future Scope)

3. Decision Justification

The selected core project pairs clinical dataset validation with a calibrated risk score to replace degenerate raw metrics, followed by interactive Tableau dashboards and a responsive Flask web app.